Internal Algorithmic Auditing and Accountability Frameworks (Raji et al. 2020)

Closing the AI accountability gap requires end-to-end internal auditing frameworks across system lifecycles, establishing verifiable documentation, ownership, and testing prior to organizational deployment.

GRADE: HIGH | STATUS: STABLE | DOMAIN: ORGANIZATIONAL, TECHNICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Voluntary ethical principles and post-hoc model explainability fail to ensure algorithmic accountability without structured, institutionalized internal audit trails. Raji et al. (2020) define the Internal Algorithmic Audit framework (SMACTR: Scoping, Mapping, Artifact Collection, Testing, and Reflection), translating high-level regulatory mandates into verifiable engineering artifacts across the entire development lifecycle.

EVIDENCE AND CAUSAL ASSESSMENT

Case studies in algorithmic bias and corporate failure reveal that harms occur primarily not from code bugs, but from organizational gaps: unrecorded assumptions, absent data lineage, and lack of defined institutional accountability for failure modes.

EFFECT SIZE AND PERSISTENCE

Pre-deployment auditing identifies systemic performance disparities and compliance violations that routine unit testing fails to capture, formalizing a paper trail necessary for regulated industry compliance (finance, healthcare, employment).

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as: robust (<20%).

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WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Completing an internal audit does not prove that an algorithm is flawless, bias-free, or mathematically incapable of future failure in unseen production environments.

WHAT ACTION IS REASONABLY JUSTIFIED

Mandate end-to-end artifact collection (data sheets, model cards, audit logs) and independent sign-off gates before deploying algorithmic systems into regulated operational environments.

VERIFIED SOURCES & DOI IDENTIFIERS

- Raji, I. D., Smart, A., White, R. N., Mitchell, M., Gebru, T., Hutchinson, B., Smith-Loud, J., Theron, D., & Barnes, P. (2020). Closing the AI accountability gap: defining an end-to-end framework for internal algorithmic auditing. Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT\ '20)*, 33–44. https://doi.org/10.1145/3351095.3372873